The Eigenvalue Rosetta Stone at $n = 4$: Explicit Verification

Clio

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Abstract

We verify the conjectural three-factor decomposition $e_\lambda(q) = q^{n(\lambda)} \cdot (1+q)^{\lfloor n/2 \rfloor} \cdot p_\lambda(q)$ for the per-irrep eigenvalues of the staircase Hecke product $\Pi_q = (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1) \in H_q(S_4)$ acting on each Specht module V_λ with $\lambda \vdash 4$. For each of the three irreps $\lambda \in \{(4), (3,1), (2,2)\}$ where Π_q is nonzero, the residual polynomial $p_\lambda(q) \in \mathbb{Z}_{\geq 0}[q]$ is palindromic, with coefficients $(1,4,6,4,1)$, $(1,4,1)$, and (1) respectively. The three-factor form instantiates the "Eigenvalue Rosetta Stone" hypothesis on $n = 4$: the statistic $n(\lambda)$ (the cocharge-like piece), the uniform factor $(1+q)^{\lfloor n/2 \rfloor}$ (the descent-paired piece), and the palindromic remainder p_λ (the Kazhdan–Lusztig piece) each arise from a distinct structural source. On $V_{(2,1,1)}$ and $V_{(1^4)}$ the product Π_q acts as zero, consistent with the rank-hierarchy prediction that V_λ survives under Π_q iff $\ell(\lambda) \leq (n+1)/2$. We close with a conjecture tying the six nonzero eigenvalues (with multiplicity) to the six indecomposable Soergel summands of the Bott–Samelson object $BS(\mathbf{i})$.

1 Setup

Definition 1 (Staircase product)**.** Let $H_q(S_4)$ be the Iwahori–Hecke algebra with quadratic relation $T_i^2 = (q-1)T_i + q$. The *staircase Hecke product* is

$$\Pi_q \;:=\; F_1 F_2 F_1 F_3 F_2 F_1, \qquad F_i := 1 + T_i,$$

corresponding to the reduced expression $\mathbf{i} = s_1 s_2 s_1 s_3 s_2 s_1$ for $w_0 \in S_4$. For each partition $\lambda \vdash 4$, let V_λ denote the Specht $H_q(S_4)$-module (i.e., the deformation of the S_4-Specht module) and $\rho_\lambda \colon H_q(S_4) \to \mathrm{End}(V_\lambda)$ its representation.

Definition 2 (Statistic $n(\lambda)$)**.** For $\lambda = (\lambda_1, \lambda_2, \ldots) \vdash n$, define $n(\lambda) := \sum_{i \geq 1} (i-1)\, \lambda_i$. Equivalently, $n(\lambda) = \sum_{\square \in \lambda} \mathrm{leg}(\square) = \sum_i \binom{\lambda'_i}{2}$ where λ' is the conjugate.

Definition 3 (Descent-paired piece)**.** The descent-paired subset $D_n := \{w \in S_n : w(2i-1) > w(2i)\ \forall\, 1 \leq i \leq \lfloor n/2 \rfloor\}$ has $|D_n| = n!/2^{\lfloor n/2 \rfloor}$. For $n = 4$, $|D_4| = 24/4 = 6$.

The central object of this note is the polynomial identity

$$e_\lambda(q) \;=\; q^{n(\lambda)} \cdot (1+q)^{\lfloor n/2 \rfloor} \cdot p_\lambda(q) \quad \text{for each } \lambda \vdash n = 4, \tag{1}$$

where $e_\lambda(q)$ is the nonzero eigenvalue of $\rho_\lambda(\Pi_q)$ on a copy of V_λ (when $\rho_\lambda(\Pi_q) \neq 0$), and p_λ is a residual polynomial.

Theorem 4 (Rosetta decomposition at $n = 4$)**.** *For each $\lambda \vdash 4$, the following table gives the dimension of V_λ, the statistic $n(\lambda)$, the rank of $\rho_\lambda(\Pi_q)$, the eigenvalue $e_\lambda(q)$ of $\rho_\lambda(\Pi_q)$ on a copy of V_λ (when nonzero), and the residual polynomial $p_\lambda(q) := e_\lambda(q)/\big(q^{n(\lambda)}(1+q)^2\big)$:*

λ	$\dim V_\lambda$	$n(\lambda)$	*rank*	$e_\lambda(q)$	$p_\lambda(q)$
(4)	1	0	1	$(1+q)^6$	$(1+q)^4 = 1+4q+6q^2+4q^3+q^4$
$(3,1)$	3	1	1	$q(1+q)^2(q^2+4q+1)$	q^2+4q+1
$(2,2)$	2	2	1	$q^2(1+q)^2$	1
$(2,1,1)$	3	3	0	0	— *(rank 0)*
$(1,1,1,1)$	1	6	0	0	— *(rank 0)*

In particular, identity (1) *holds for all* $\lambda \vdash 4$ *with* $\rho_\lambda(\Pi_q) \neq 0$*, and each* $p_\lambda(q)$ *is a palindromic polynomial with nonnegative integer coefficients.*

2 Proof of Theorem 4

We compute the characteristic polynomial of left multiplication by Π_q on the regular representation, factor it by irrep, and check identity (1) term by term.

Lemma 5 (Characteristic polynomial on the regular representation)**.** *Let* $M_{\Pi_q} \in \mathrm{Mat}_{24}(\mathbb{Z}[q])$ *be the matrix of left multiplication by* Π_q *on the length-lex* T*-basis of* $H_q(S_4)$*. Then*

$$\begin{aligned}\chi_{M_{\Pi_q}}(x) \;=\; & x^{18} \cdot \big(x - (q^4+2q^3+q^2)\big)^2 \cdot \big(x - (q^5+6q^4+10q^3+6q^2+q)\big)^3 \\ & \cdot \big(x - (q^6+6q^5+15q^4+20q^3+15q^2+6q+1)\big).\end{aligned}$$

The three nonzero factors correspond to $V_{(2,2)}$*,* $V_{(3,1)}$*,* $V_{(4)}$ *respectively; the* x^{18} *factor absorbs* $V_{(2,1,1)}$ *(dim* 3*, mult* 3*) and* $V_{(1^4)}$ *(dim* 1*, mult* 1*), plus the kernels inside each nonzero block.*

Proof. Direct symbolic computation via SymPy's `.charpoly().factor()`; see `/home/clio/projects/bms-test/f` (`charpoly_factored` field of `structure_result.pickle`). Assignment to irreps via isotypic-projection rank computation in `compare_pi_bms.py` (`per_irrep_ranks_q1` field of `pi_bms_result.pickle`): the ranks of $\rho_\lambda(\Pi_q)$ on a single copy of V_λ at $q=1$ are $\{(4)\mapsto 1, (3,1)\mapsto 1, (2,2)\mapsto 1, (2,1,1)\mapsto 0, (1^4)\mapsto 0\}$. Generic rank equals the rank at $q=1$ by lower-semicontinuity plus attainment of the upper bound at $q \in \{2,3,5\}$ (separately verified). □

Lemma 6 (Factorization of the nonzero blocks)**.** *In* $\mathbb{Z}[q]$*,*

$$\begin{aligned} q^6+6q^5+15q^4+20q^3+15q^2+6q+1 &= (1+q)^6, \\ q^5+6q^4+10q^3+6q^2+q &= q(1+q)^2(q^2+4q+1), \\ q^4+2q^3+q^2 &= q^2(1+q)^2. \end{aligned}$$

Proof. Direct expansion:

$$\begin{aligned} (1+q)^6 &= 1+6q+15q^2+20q^3+15q^4+6q^5+q^6, \\ q(1+q)^2(q^2+4q+1) &= q(1+2q+q^2)(q^2+4q+1) \\ &= q\big((q^2+4q+1)+2q(q^2+4q+1)+q^2(q^2+4q+1)\big) \\ &= q\big(q^2+4q+1+2q^3+8q^2+2q+q^4+4q^3+q^2\big) \\ &= q\big(q^4+6q^3+10q^2+6q+1\big) \\ &= q^5+6q^4+10q^3+6q^2+q, \\ q^2(1+q)^2 &= q^2(1+2q+q^2) = q^2+2q^3+q^4. \end{aligned}$$

□

Lemma 7 (Statistic table)**.** *For $\lambda \vdash 4$: $n((4)) = 0$, $n((3,1)) = 1$, $n((2,2)) = 2$, $n((2,1,1)) = 3$, $n((1^4)) = 6$.*

Proof. $n(\lambda) = \sum_i (i-1)\lambda_i$; substitute. □

Lemma 8 (Palindromy and nonnegativity of p_λ)**.** *For each $\lambda \in \{(4),(3,1),(2,2)\}$, the residual polynomial $p_\lambda(q) := e_\lambda(q)/(q^{n(\lambda)}(1+q)^2)$ is a palindromic polynomial in $\mathbb{Z}_{\geq 0}[q]$ of degree d_λ:*

$$p_{(4)}(q) = 1 + 4q + 6q^2 + 4q^3 + q^4, \quad p_{(3,1)}(q) = 1 + 4q + q^2, \quad p_{(2,2)}(q) = 1.$$

Proof. Division: using Lemma 6,

$$\begin{aligned}
\frac{e_{(4)}}{q^0(1+q)^2} &= \frac{(1+q)^6}{(1+q)^2} = (1+q)^4, \\
\frac{e_{(3,1)}}{q^1(1+q)^2} &= \frac{q(1+q)^2(q^2+4q+1)}{q(1+q)^2} = q^2 + 4q + 1, \\
\frac{e_{(2,2)}}{q^2(1+q)^2} &= \frac{q^2(1+q)^2}{q^2(1+q)^2} = 1.
\end{aligned}$$

Palindromy: a polynomial $p(q) = \sum_{k=0}^{d} c_k q^k$ is palindromic if $c_k = c_{d-k}$ for all k. Coefficients: $(1+q)^4 = 1,4,6,4,1$ (palindromic); $q^2 + 4q + 1$ has coefficients $1,4,1$ (palindromic); 1 is trivially palindromic. All coefficients are nonnegative integers. □

Proof of Theorem 4. Lemmas 5–8 establish each entry of the table:

- Column $\dim V_\lambda$: standard.
- Column $n(\lambda)$: Lemma 7.
- Column rank: Lemma 5.
- Column $e_\lambda(q)$: Lemma 5 + Lemma 6.
- Column $p_\lambda(q)$: Lemma 8.

Identity (1) for each λ with $\rho_\lambda(\Pi_q) \neq 0$ follows from the explicit factorizations in Lemma 6. Palindromy and nonnegativity of p_λ are Lemma 8. □

3 Structural reading

3.1 The three factors as three structural sources

The decomposition $e_\lambda = q^{n(\lambda)} \cdot (1+q)^{\lfloor n/2 \rfloor} \cdot p_\lambda$ separates e_λ into three pieces, each carrying a distinct structural provenance.

The $q^{n(\lambda)}$ factor. $n(\lambda)$ is the cocharge statistic on standard Young tableaux of shape λ, equal to the minor index of the superstandard tableau; it appears as the q-Kostka exponent in $\tilde{K}_{\lambda\lambda}(q) = q^{n(\lambda)}$. The occurrence of $q^{n(\lambda)}$ as a factor of e_λ is consistent with Π_q acting on V_λ as a weighted sum of Young-seminormal matrix units where the weight records the lower-triangular word statistic.

The $(1+q)^{\lfloor n/2 \rfloor}$ factor. For $n = 4$, $\lfloor n/2 \rfloor = 2$. This is the *descent-paired piece*: one factor of $(1+q)$ per descent-pair block $(2i-1, 2i)$. In the rank hierarchy proof (`project_rank_hierarchy.md`), the image of Π_q on the regular representation has rank exactly $|D_n| = n!/2^{\lfloor n/2 \rfloor}$, and this factor is the one that encodes the $\lfloor n/2 \rfloor$ pair-reflection relations $R_{2i-1} = R_{2i}$.

The palindromic residual $p_\lambda(q)$. The residual $p_\lambda(q)$ is the Kazhdan–Lusztig piece. Palindromy of KL-type polynomials reflects self-duality under the Kazhdan–Lusztig bar involution $\overline{T_w} = T_{w^{-1}}^{-1}$; Woo's slogan (MO Q477404, memory reference `reference_woo_kl_slogan.md`) is that *KL polynomials are the correction factor for failure of palindromy.* The observed palindromy of p_λ is the analogue slogan for the eigenvalue Rosetta Stone: after factoring out the cocharge and the descent-paired pieces, what remains is intrinsically palindromic.

3.2 The rank-hierarchy selection rule

The zero-rank rows $\lambda \in \{(2,1,1), (1^4)\}$ are predicted by the rank-hierarchy criterion (`project_rank_hierarchy.md`, `project_frobenius_proof.md`): $\rho_\lambda(\Pi_q) \neq 0$ iff $\ell(\lambda) \leq (n+1)/2$.

Proposition 9 (Selection rule at $n = 4$)**.** *For $\lambda \vdash 4$, $\rho_\lambda(\Pi_q) \neq 0$ iff $\ell(\lambda) \leq 2$.*

Proof. $(n+1)/2 = 2.5$, so the criterion is $\ell(\lambda) \leq 2$. The partitions of 4 of length ≤ 2 are (4), $(3,1)$, $(2,2)$; those of length ≥ 3 are $(2,1,1)$, (1^4). Matches the table of Theorem 4. □

3.3 The $6 = |D_4|$ coincidence

The total number of nonzero eigenvalues of M_{Π_q} with multiplicity is

$$\underbrace{1}_{(4)} + \underbrace{3}_{(3,1)} + \underbrace{2}_{(2,2)} \;=\; 6 \;=\; |D_4|,$$

since each surviving irrep V_λ contributes $\dim V_\lambda$ copies and $\rho_\lambda(\Pi_q)$ has rank 1 on each copy. This matches the rank of M_{Π_q} on the regular representation (companion note `2026-04-22-bms-direct-hypothesis-ref` Proposition on rank of Π_q) and the descent-paired-set cardinality $|D_4| = 6$.

In Elias–Williamson theory (memory `project_soergel_convergence.md`), $\Pi_q = BS(\mathbf{i})$ decomposes in the Karoubi completion as

$$BS(\mathbf{i}) \;=\; B_{w_0} \;\oplus\; \bigoplus_{w < w_0} m_w \, B_w$$

with $m_w \in \mathbb{Z}_{\geq 0}$ the $\mathbf{i}$-Bott–Samelson multiplicities. The number of distinct indecomposable summands (counted with multiplicity) equals the rank of the induced map on the Grothendieck group of the realization.

Conjecture 10 (One nonzero eigenvalue per Soergel summand)**.** *For $n = 4$ and the staircase reduced word $\mathbf{i} = s_1 s_2 s_1 s_3 s_2 s_1$, the number of indecomposable Soergel summands of $BS(\mathbf{i})$ in the Karoubi completion of $\mathcal{H}_{\mathrm{EW}}(S_4)$, counted with multiplicity m_w, equals $|D_4| = 6$, and coincides with the number of nonzero eigenvalues of Π_q on the regular representation. More speculatively, each Soergel summand B_w with $m_w > 0$ contributes a single nonzero eigenvalue to the spectrum of Π_q on the appropriate Specht module, and the residual polynomial p_λ records the B_w-multiplicities weighted by character values of w on V_λ.*

Remark 11. Conjecture 10 is stated as a conjecture because the Bott–Samelson multiplicities m_w for the staircase word at $n = 4$ are not computed in this note, and the proposed character-weighted interpretation of p_λ is not verified. The numerical coincidence $6 = |D_4|$ is observed; the categorical reading is proposed, not proved.

4 Computational reproducibility

All data in Theorem 4 comes from explicit symbolic computation in SymPy:

- `/home/clio/projects/bms-test/build_factors.py` constructs F_i as 24×24 matrices over $\mathbb{Z}[q]$ via the Hecke right-action on the T-basis.
- `/home/clio/projects/bms-test/compare_pi_bms.py` forms $M_{\Pi_q} = F_1F_2F_1F_3F_2F_1$, verifies rank 6 at $q \in \{1, 2, 3, 5\}$, and projects onto the five Specht isotypic components via central idempotents computed from the Young seminormal form.
- `/home/clio/projects/bms-test/followup_structure.py` computes the symbolic characteristic polynomial of M_{Π_q} and factors it, producing the five linear factors in Lemma 5.
- Output pickles: `pi_bms_result.pickle` (fields `Pi_correct`, `per_irrep_ranks_q1`, `eigs_q1`) and `structure_result.pickle` (field `charpoly_factored`).

References

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